Findings: Caveats and a Reading Guide

> Part of the Functional Series. This document closes the series. It gathers the uncertainties that the findings carry forward, says what the platform cannot conclude, and gives a short guide for quoting its results responsibly.

1. Uncertainties carried forward

These are open, not resolved. Each is stated where it first arose.

Uncertainty	Why it is open	Effect on the findings
China's trajectory	The platform's BAU keeps rising to 2030 (about +10%); UNEP's policy-based studies show a peak around 2025	The platform's forecast for China, and so for the global total, is likely too high; it is a continuation of past behaviour, not a policy projection (Finding 12)
Methane	The platform's methane (8.2 $GtCO_2e$ in 2019) is below the IPCC's (11) and UNEP's (9.3 for 2024); the cause was not traced	The platform's total greenhouse gas figure is lower than global inventories partly for an unexplained reason (Emissions §2)
Cement carbonation sink	The platform's 2024 fossil CO_2 (38.6) is about 0.7 $GtCO_2$ above the Global Carbon Budget (37.8), the size of the sink the latter nets out; OWID's codebook does not state its treatment	The explanation is consistent but not verified; budget crossing years come out slightly early (Targets §4)
Preliminary temperature data	Berkeley Earth labels its high-resolution product "preliminary", and the platform's recent years differ from Berkeley's own report by up to about 0.03 °C	The latest single-year temperatures may be revised
Two carbon budgets	130 and 170 $GtCO_2$ for 1.5 °C, from different assessments	Crossing years move by at most a year
Land use	About ±2.6 $GtCO_2$ a year (one standard deviation) and held flat in scenarios	Totals including land use are the least certain figures in the series
Source texts	The UNFCCC's own decision documents could not be retrieved for this series	The wording of the COP28 and COP30 decisions should be checked against the official text before it is quoted

2. What the platform cannot conclude

- **Policy effects.** It does not model policy, prices, technology or behaviour. It cannot say what any pledge or law will do.
- **Temperature beyond 2040.** The horizon is 2040; the published outlooks to 2100 are other organisations' work.
- **Attribution to countries.** No country series is regressed against temperature, and no warming is assigned to a country.

- **Causation.** The emissions–temperature relationship is a statistical association over 175 years, offered as context.
- **Whether any single pledge is fair or ambitious.** That needs a common metric the platform does not compute.
- **Feasibility.** That a pathway requires a given rate of cuts does not show it can or cannot be achieved.

3. How to quote this work

Do	Do not
Say which source, edition and scope a number comes from	Quote a total without saying whether it includes land use, bunkers, or all gases
Describe scenarios as *illustrative*	Describe the 2040 temperature levels as forecasts
Say "likely range" for the IPCC's TCRE (66%+), and "95% interval" for the platform's slope	Treat the two as the same confidence level
Compare the platform's trend with a benchmark on the same scope	Compare a gross figure with a net pledge and call the gap a failure
Use multi-year averages for the Paris temperature goal	Say the 1.5 °C limit has been "breached" on the strength of one or two years
Date every statement about pledges and the COP process	Present late-2025 facts as current

4. One-paragraph summary

Emissions have not yet fallen from their 2019 level, and atmospheric CO_2 and temperature continue to rise on the paths the IPCC describes. Long-term warming is about 1.4 °C, and the remaining 1.5 °C carbon budget is almost spent: the platform's scenarios, which are illustrative, all use it up by about 2030 and none follows a Paris-aligned pathway. The required rate of cuts (about 4.6% a year for 2 °C, about 10% for 1.5 °C) has no precedent in the record since 1960. National pledges for 2035 would cut emissions about 12–15% from 2019, against the 35–55% that pathways require, and published warming outlooks for 2100 range from about 1.9 to 3.2 °C depending on what is implemented. Countries differ widely: some peaked decades ago, while China and India are at record highs, and China is where the platform's statistical forecast is least likely to capture policy-driven change. Every one of these statements has limits, set out above and in the sections they come from.

See also

- Where Things Stand
- Trajectory Against Paris
- Country Differences
- Limitations and Uncertainty